

Synchronization - Part 2

Operating Systems – EDA093/DIT401

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What to read (Main textbook)

- Both for part 1 and part 2

- 4th edition

- Chapter 2.3.1-2.3.6, 2.5.1-2.5.2, 6.1-6.2, 6.5-6.6, 6.7.3-6.7.4;
- Quicker reading, for awareness, of sections 2.3.7-2.3.10, 6.3

- 5th edition

- Chapter 2.4.1-2.4.6, 6.1-6.2, 6.5-6.6, 6.7.3-6.7.4;
- Quicker reading, for awareness, of sections 2.3.7-2.3.9, 2.3.11, 6.3

(facultative reading: sections 6.1-6.7, 6.9, 7.1-7.5, 7.6-7.8 from OS Concepts by Silberschatz et-al).

Agenda

- Bounded-buffer producer/consumer
- Resource allocation, deadlocks, and necessary conditions for deadlocks
- Dining philosophers
 - without circular wait
 - without no-preemption
 - without hold-and-wait
- Lamport's bakery algorithm
- Readers/Writers problem
- OS as arbitrator for deadlock avoidance/recovery

Agenda

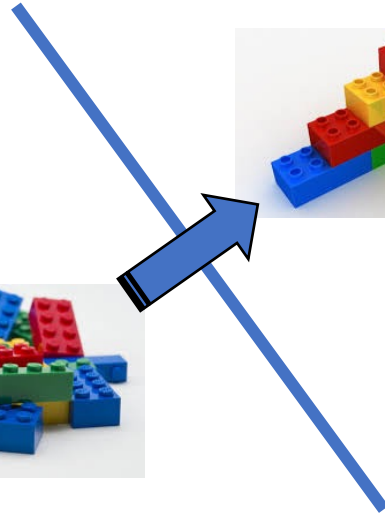
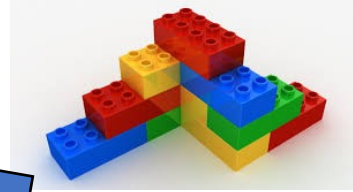
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the Bounded buffer producer-consumer problem

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Using: primitives

- R/W variables
- RMW variables
- Semaphores, etc.
- ...

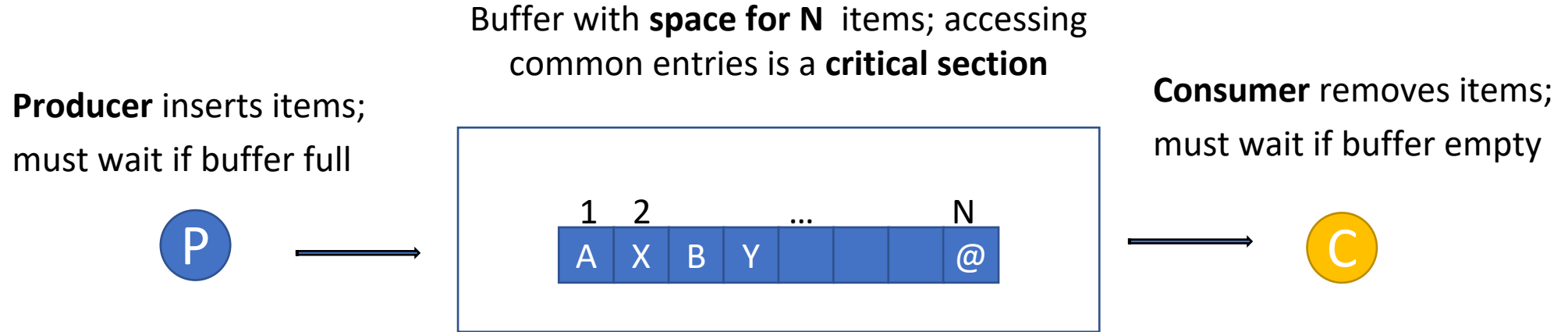


Construct: objects / solve specific synchronization problems

- 2 thread CS, n-thread-CS
- Producer-consumer (bounded buffer)
- Dining philosophers
- Transactions
- ...

the Bounded buffer producer-consumer problem

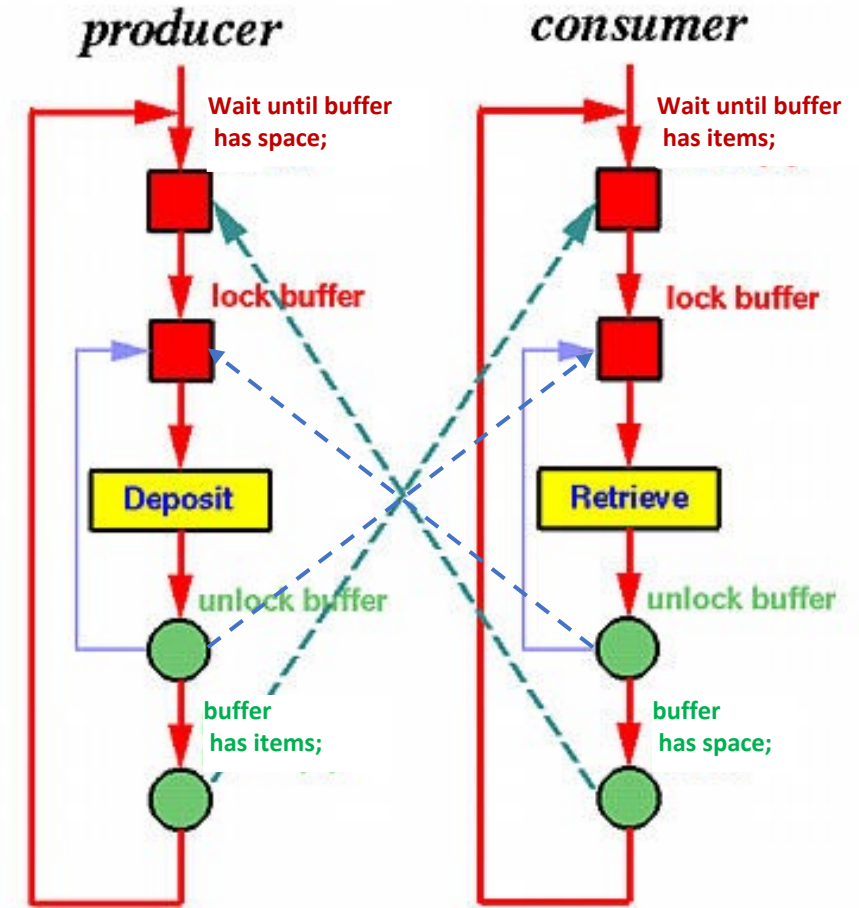
6



Solve this synch problem using semaphores

What synchronization do we need?

- **Producer** inserts items; must wait if buffer full
- **Consumer** removes items; must wait if buffer empty
- Accessing the buffer is a **critical section**



a solution to the Bounded buffer producer-consumer problem 8

Synchronization variables:

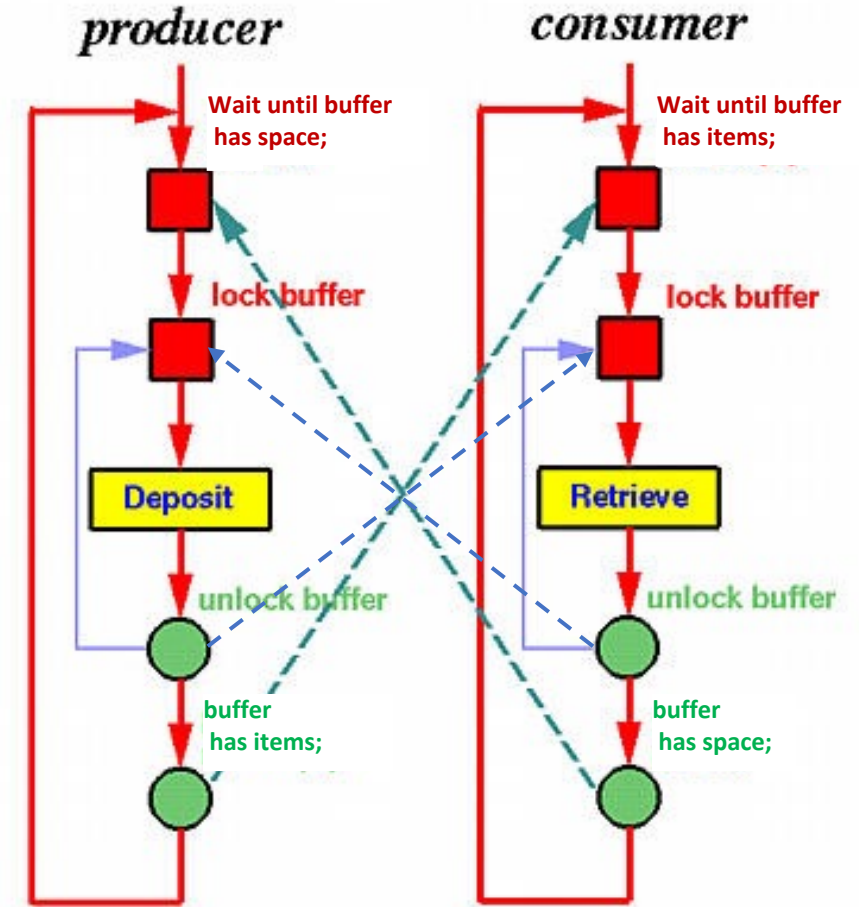
- Binary semaphore **mutex_sem** initialized to 1
- General semaphore **buffer-has-items** initialized to 0
- General semaphore **buffer-has-space** initialized to N

producer

```
do {  
    // produce item  
    wait(buffer-has-space);  
    wait(mutex_sem);  
    // add item to buffer  
    signal(mutex_sem);  
    signal(buffer-has-items);  
} while (TRUE);
```

consumer

```
do {  
    wait(buffer-has-items);  
    wait(mutex_sem);  
    // remove item from buffer  
    signal(mutex_sem);  
    signal(buffer-has-space);  
    // use the item  
} while (TRUE);
```



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Resource allocation

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- Processes/threads need resources (e.g., memory pages, printer, access to parts of shared data structure, etc.)
 - Our focus: reusable resources
- a human analogy: process = go fishing; needed resources: boat, fishing-rod



Structure of process/thread P

Repeat

Request resources (entry section)

critical section

Release resources (exit section)

remainder section

Forever

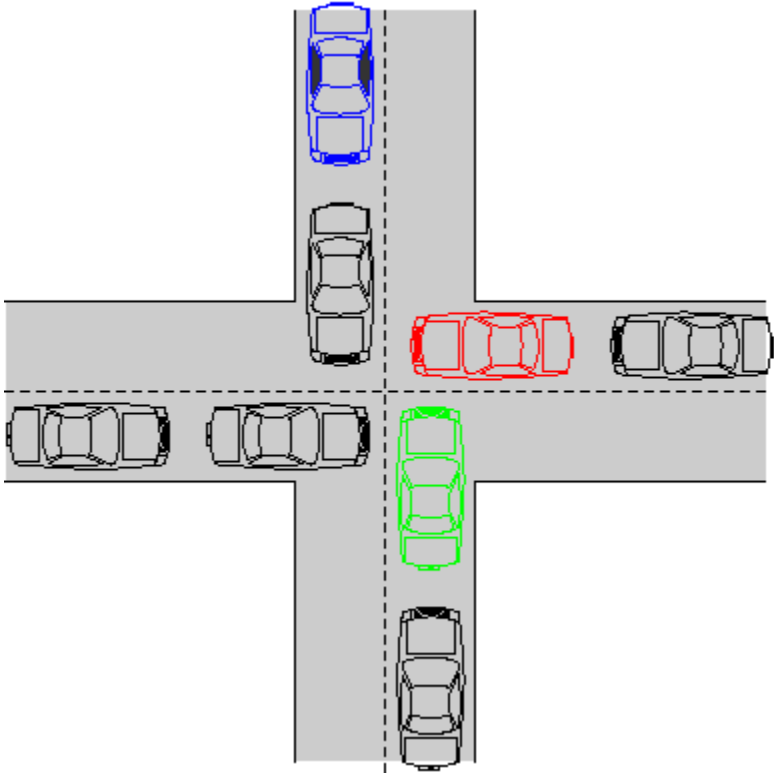
Problem formulation:

A solution must **provide** the **entry** and **exit** sections

It must **ensure**:

- 1. Acquire/Release all the needed resources**
- 2. Mutual Exclusion.**
- 3. Progress: no deadlock**
- 4. Fairness (e.g., Bounded Waiting , no starvation)**

What is a deadlock?

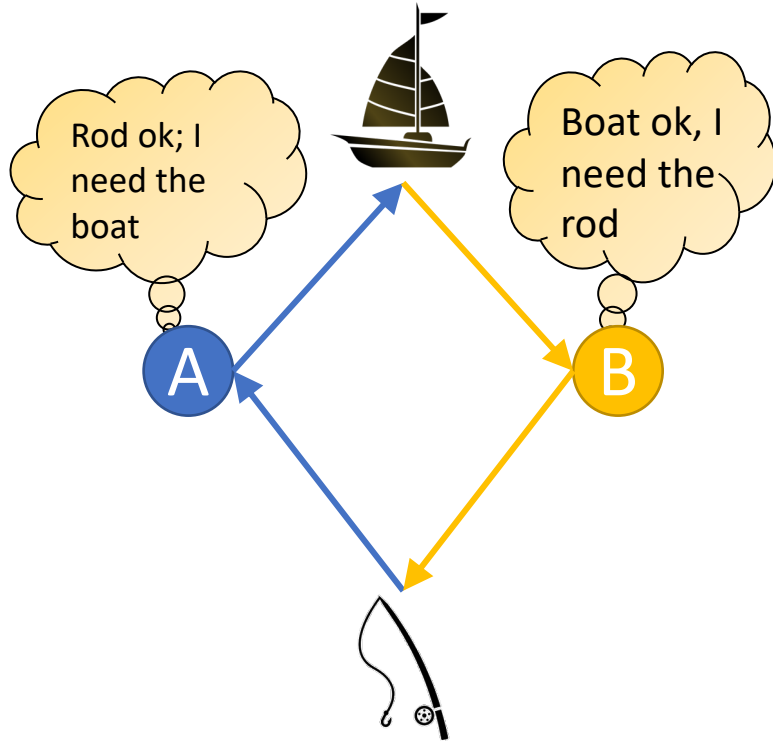


A set of processes/threads blocking each-other so that none of them can proceed:

How can it occur?

4 necessary conditions for Deadlock [Coffman et al 1971]

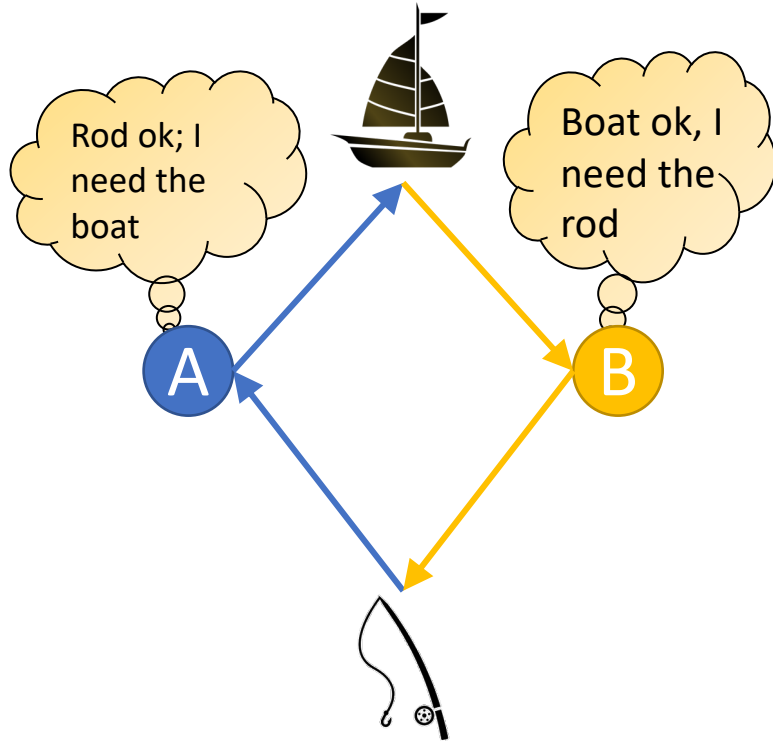
Theorem: all 4 conditions hold simultaneously when a deadlock occurs:



1. **Mutual exclusion:** only one process at a time can use a resource.
2. **Hold and wait:** a process holding some resource can request additional resources and wait for them if they are held by other processes.
3. **No preemption:** a resource can only be released by the process holding it, after that process has completed its task.
4. **Circular wait:** there exists a circular chain of 2 or more blocked processes, each waiting for a resource held by the next proc. in the chain

4 necessary conditions for Deadlock [Coffman et al 1971]

Theorem: all 4 conditions hold simultaneously when a deadlock occurs:



1. **Mutual exclusion:** only one process at a time can use a resource.
2. **Hold and wait:** a process holding some resource can request additional resources and wait for them if they are held by other processes. → **GET ALL RESOURCES AT ONCE**
3. **No preemption:** a resource can only be released by the process holding it, after that process has completed its task. → **THREADS RELEASE RESOURCES IF THEY MANAGE TO GET SOME BUT NOT ALL**
4. **Circular wait:** there exists a circular chain of 2 or more blocked processes, each waiting for a resource held by the next proc. in the chain. → **ACQUIRE RESOURCES IN A CERTAIN ORDER**

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Dining philosophers [Dijkstra65]

n philosophers (processes); each philosopher P_i , when hungry, needs both left & right chopstick, in order to eat

Structure of process/thread P

Repeat

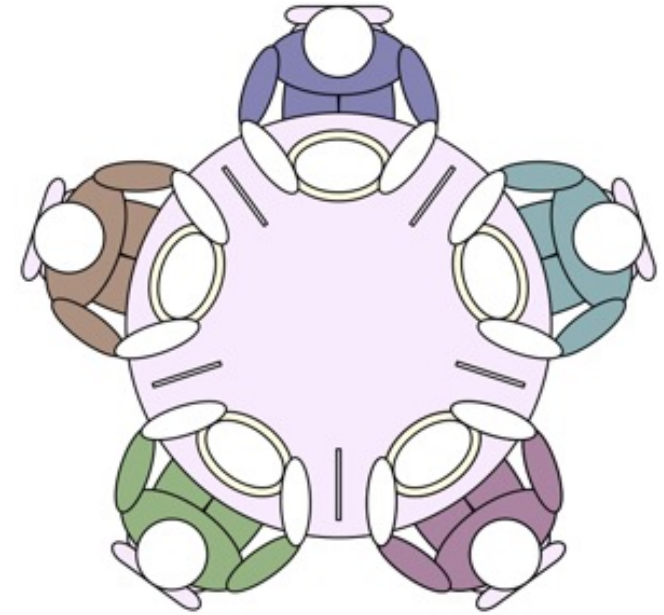
Request resources (left + right chopstick)

eat

Release resources (left + right chopstick)

think

Forever



Dining philosophers: pick-left-then-right approach

```
1.  Shared var c[0..n-1]: bin-semaphore // one for
    each chopstick; init all 1
2.  Pi:
3.  do
4.      Wait c[i]; // pick left chopstick
5.      Wait c[(i+1) mod n]; // pick right chopstick
6.      // Eat
7.      Signal c[i]; // leave left chopstick
8.      Signal c[(i+1) mod n]; leave right chopstick
9.      // Think
10. forever
```

Recall the requirements:

1. **Mutual exclusion:** each resource is used by only one process at a time
2. **Progress:** no deadlock
3. **Fairness:** FCFS, or no starvation, or other fairness formulation

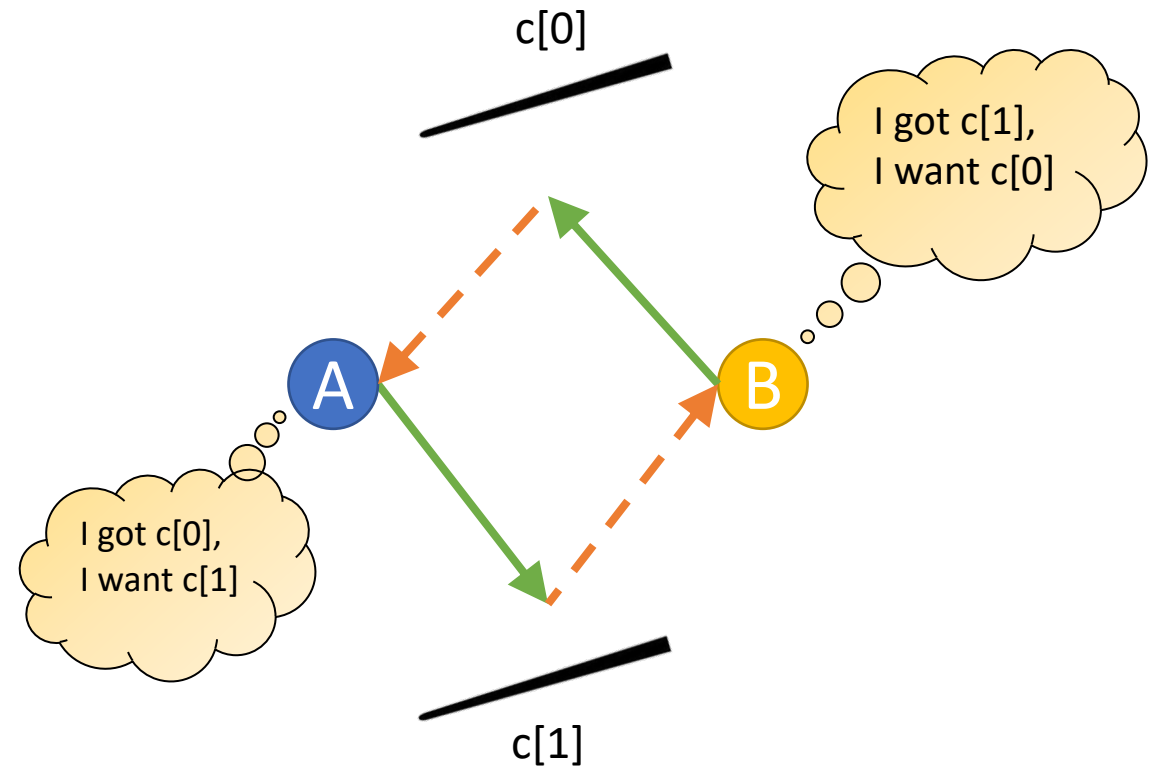
Dining philosophers: pick-left-then-right approach

```
1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2. Pi:
3. do Hold and wait
4.   Wait c[i]; // pick left chopstick
5.   Wait c[(i+1) mod n]; // pick right chopstick
6.   // Eat
7.   Signal c[i]; // leave left chopstick
8.   Signal c[(i+1) mod n]; // leave right chopstick
9.   // Think No preemption
10. forever
```

Dining philosophers: pick-left-then-right approach

```

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   each chopstick; init all 1
2. Pi:
3. do                                     Hold and wait
4.   Wait c[i]; // pick left chopstick
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6.   // Eat
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9.   // Think                             No preemption
10. forever
  
```



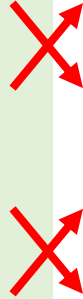
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Dining philosophers: pick-one-at-a-time without circular wait

```

1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2.  $P_i$  (except  $P_{n-1}$ ):
3. do
4.     Wait c[i]; // pick left chopstick
5.     Wait c[(i+1) mod n]; // pick right chopstick
6.     // Eat
7.     Signal c[(i+1) mod n]; leave right chopstick
8.     Signal c[i]; // leave left chopstick
9.     // Think
10. forever
  
```



```

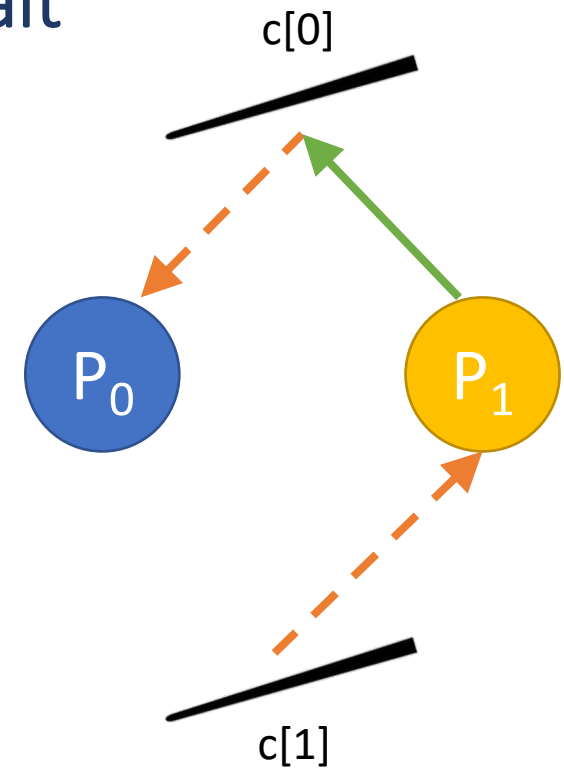
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2.  $P_{n-1}$ :
3. do
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8.     Signal c[(i+1) mod n]; leave right chopstick
9.     // Think
10. forever
  
```

Dining philosophers: pick-one-at-a-time without circular wait

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```
1.  Shared var c[0..n-1]: bin-semaphore // one
    for each chopstick; init all 1
2.  Pi (except Pn-1):
3.  do
4.    Wait c[i]; // pick left chopstick
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6.    // Eat
7.    Signal c[(i+1) mod n]; leave right chopstick
8.    Signal c[i]; // leave left chopstick
9.    // Think
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```

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1.  Shared var c[0..n-1]: bin-semaphore // one
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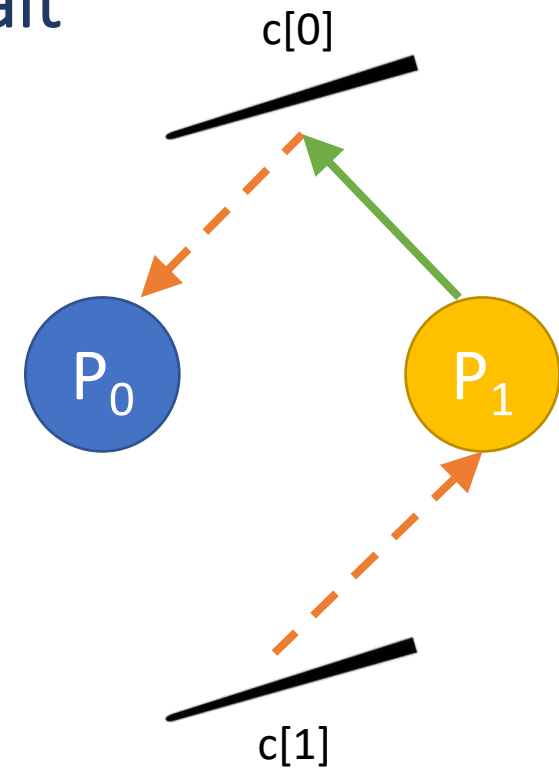
Dining philosophers: pick-one-at-a-time without circular wait

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9.    // Think
10. forever
  
```

```

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    for each chopstick; init all 1
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Dining philosophers: Fight the no-preemption

Idea: when the second resource is not available, release the first one and retry



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```
shared var c[0..n-1]: of type chopstick_struct { // one for each chopstick
  s: bin-semaphore // init 1
  available: boolean //init true
}
```

```
Pi:
  local var holding_both_chopsticks: boolean;
  repeat
    while (not holding_both_chopsticks) {
      lock(c[i])
      if !trylock(c[(i+1)%n]) then release(c[i])
      else holding_both_chopsticks := true }
    // Eat
    release(c[i])
    release(c[(i+1)%n])
    holding_both_chopsticks := false
    // Think
  forever
```

trylock(c: chopstick_struct):

```
wait(c.s)
if c.available then {
  c.available := false
  ret:= true }
else
  ret:= false;
signal(c.s)
return(ret)
```

lock(c : chopstick_struct):

```
repeat
until (trylock(c))
```

release(c : chopstick_struct):

```
wait(c.s)
c.available := true
signal(c.s)
```


Dining philosophers: Fight the no-preemption

- Mutual exclusion: ok
- Progress: no deadlock ... but livelock is possible!
- Fairness: a process can starve...

```
Pi:
  local var holding_both_chopsticks: boolean;
  repeat
    while (not holding_both_chopsticks) {
      lock(c[i])
      if !trylock(c[(i+1)%n]) then release(c[i])
      else holding_both_chopsticks := true }
    // Eat
    release(c[i])
    release(c[(i+1)%n])
    holding_both_chopsticks := false
    // Think
  forever
```

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Dining philosophers: Fight the hold-and-wait

Idea: philosophers agree on who eats next instead of who gets a chopstick



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```
shared var semaphore S[0 .. n-1] // init all 0
shared var semaphore mutex // init 1
shared var state[0 .. n-1] in {HUNGRY, THINKING, EATING}
Pi:
do
  // think
  enterCS(i) // i.e., get both chopsticks
  // eat
  exitCS(i) // i.e., leave chopsticks
forever
```

```
help(k)
if state[k] == HUNGRY &&
  state[(k-1) mod n] != EATING &&
  state[(k+1) mod n] != EATING
then {state(k) := EATING ; signal(S[k]) }
```

```
enterCS(i)
  wait(mutex)
  state(i) := HUNGRY
  help(i)
  signal(mutex)
  wait(S[i])
```

```
exitCS(i)
  wait(mutex)
  state(i) := THINKING
  help((i-1) mod n)
  help((i+1) mod n)
  signal(mutex)
```

Dining philosophers: Fight the no-preemption

- Mutual exclusion: ok
- Progress: no deadlock
- Fairness: a process can starve...
- If you are interested in a solution with fairness too:
https://dl.acm.org/doi/pdf/10.1145/62546.62567?casa_token=4uO24jxkwEAAAAAA:jJlAlLeISZe5Uu2ERv6O-dTq_0LbSmRpv0beOZ_3vDi50otRS-HqB30GoWDib1zVQ9jjrhx4w0

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Lamport's bakery algorithm

Critical section for n threads:

Idea: before entering its critical section, each thread gets a number. Holder of the smallest number enters the critical section.



Note: the decentralized scheme generates numbers in non-decreasing order; e.g., 1, 2, 3, 3, 4, 5, 5, 5, 6

If threads P_i and P_j choose the same number:

if $i < j$, then P_i goes first; else P_j goes first.

I.e., we use threads' ids to break ties

Lamport's bakery algorithm

```

Shared var choosing: array [0..n - 1] of boolean (init false);
           number: array [0..n - 1] of integer (init 0);

repeat
  choosing[i] := true;
  number[i] := max(number[0], number[1], ..., number [n - 1])+1;
  choosing[i] := false;
  for j := 0 to n - 1 do begin
    while choosing[j] do [nothing]; // spin
    while number[j] ≠ 0 and (number[j],j) < (number[i], i) do [nothing]; // spin
  end;
  // critical section
  number[i] := 0;
  // remainder section
until false;

```

Why does it satisfy the 3 conditions:

Mutex (no 2 threads A and B in CS concurrently): Consider the time between A's decision step and A's entry to CS; A decided to move because:

- B had higher number: when B checks , it will wait for A since A has smaller number
- or B was not interested; when B gets interested, it will choose a number > A's number, hence it will wait

Progress: the thread with the smaller number can proceed

Fairness: If A waits for B and B exits and wants to enter CS again, if A still waits, B will choose a number > A's, number, hence B cannot bypass A

*This is a more **decentralized** method than e.g., Peterson's: no variable is "writable" by all threads*

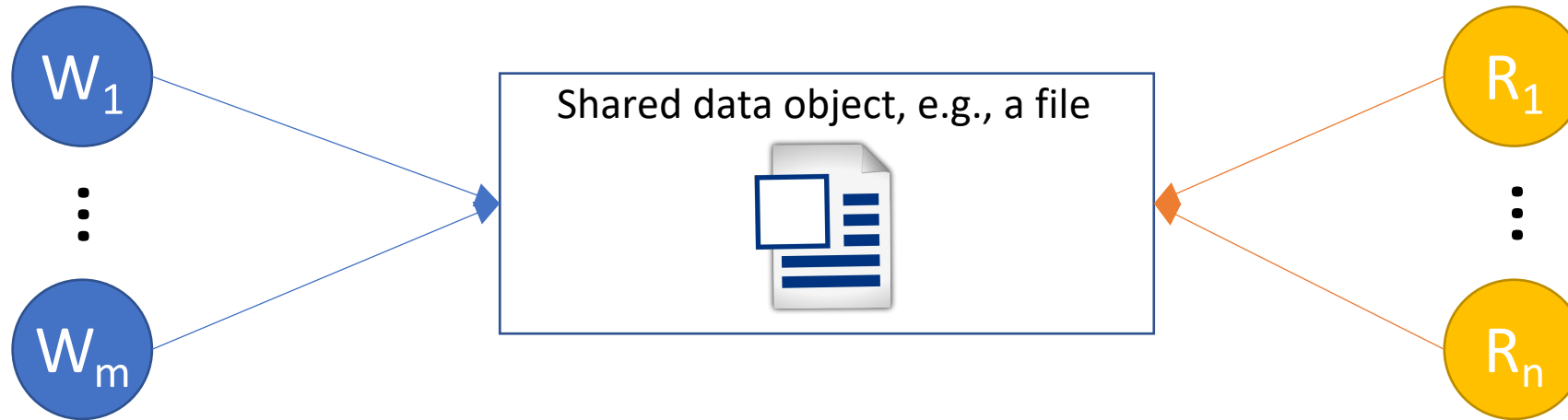
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Readers/Writers problem

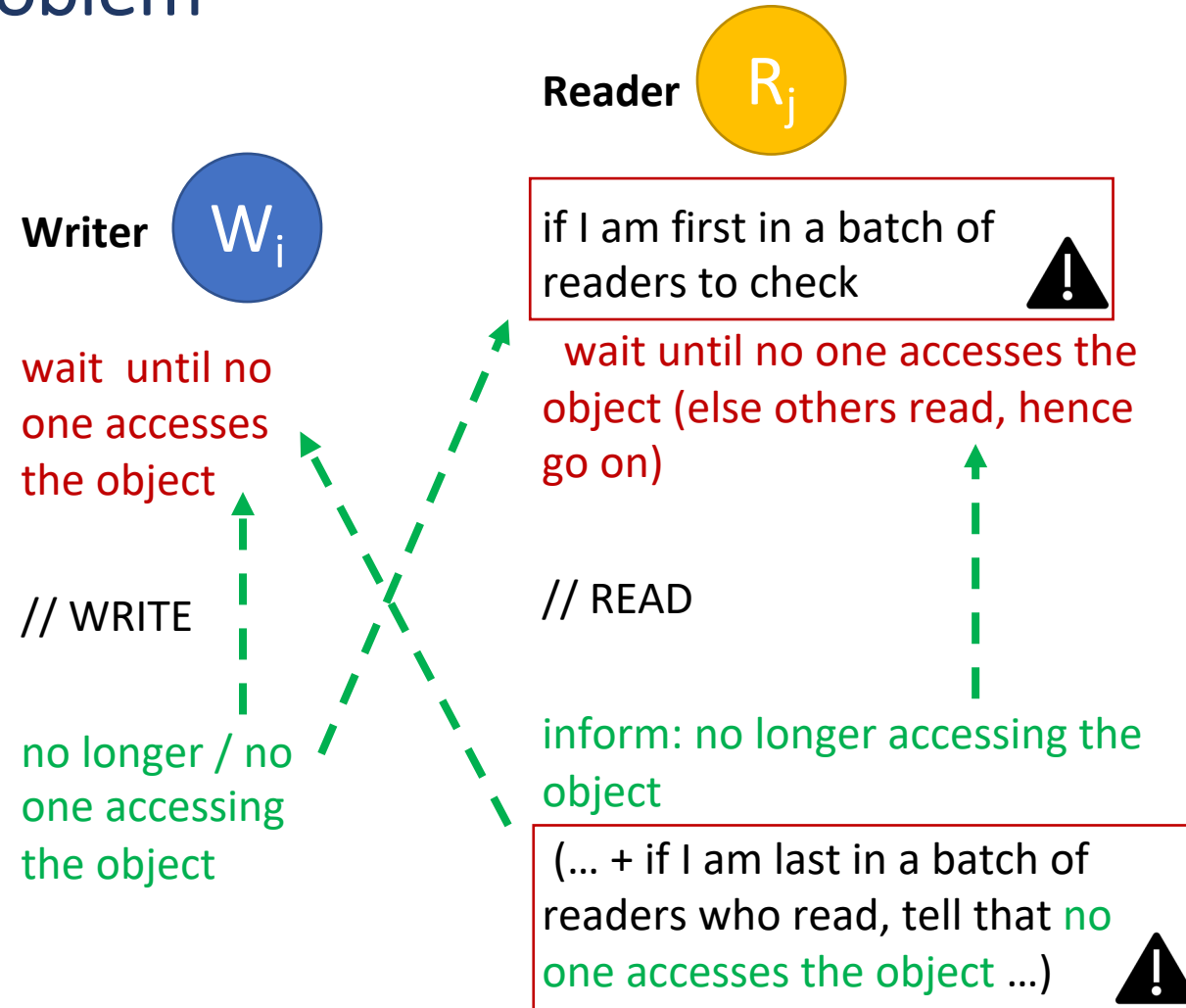
Writers write; must wait if some writer or reader is accessing the shared data object (i.e., *only one writer is allowed to write at a time*)

Readers read; must wait if some writer is accessing the shared data object (i.e., *multiple readers are allowed to read at a time but not concurrently with any writer*)



Solve this synch problem using semaphores; we require 1. the safety properties mentioned here; 2. progress

Readers/Writers problem



Readers/Writers problem

shared var:

noone_accesses, protect_check: binary semaphore; // initially 1
rc: int ; // active readers counter, init 0



Writer



Repeat

```
wait(noone_accesses);
// WRITE
signal(noone_accesses)
forever
```

Reader



repeat

```
wait(protect_check); // CS to change and check rc variable
if rc++ == 1 then wait(noone_accesses) // "first" reader: block
writers or wait if some of them writes
signal(protect_check);
// READ
wait(protect_check); // CS to change and check rc
if rc-- == 0 then signal(noone_accesses) fi // "last" reader: signals
signal(protect_check)
forever
```

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Deadlock avoidance using the OS as arbitrator

Resource request & allocation is managed by the OS

Deadlock *avoidance*:

- deadlock *might be* possible if resources are granted arbitrarily,
- but OS uses extra info to grant requests and schedule processes so that it avoids deadlock
- **Banker's algorithm[Dijkstra]**

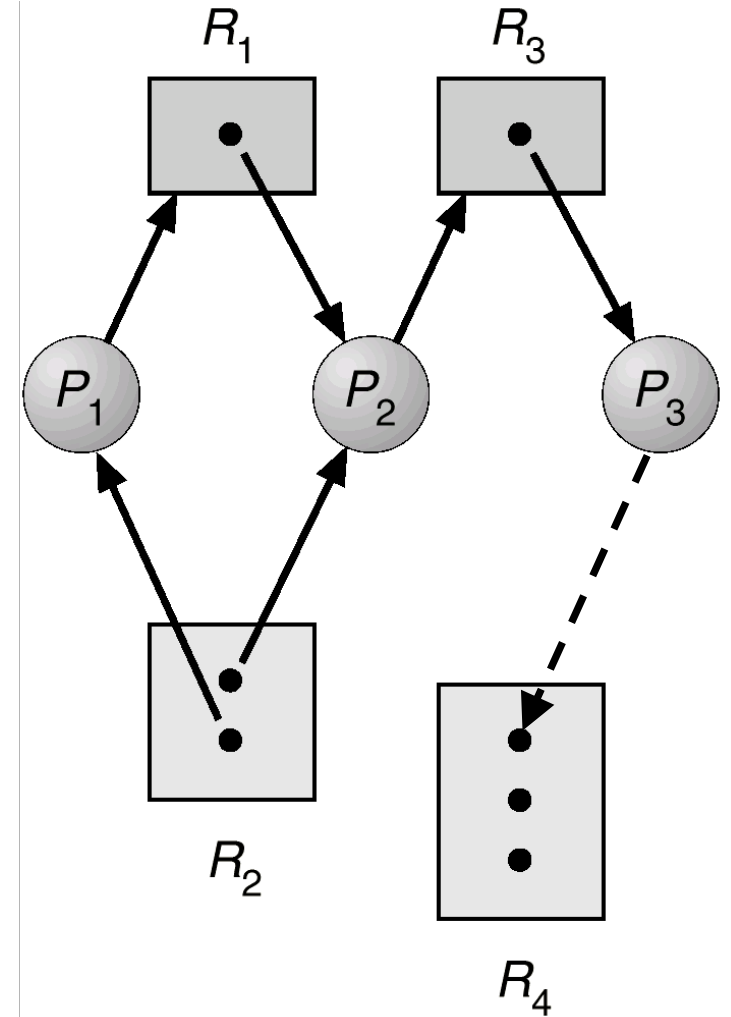
Deadlock avoidance by the OS: system model

Resource types $R_1, R_2, \dots, R_m$

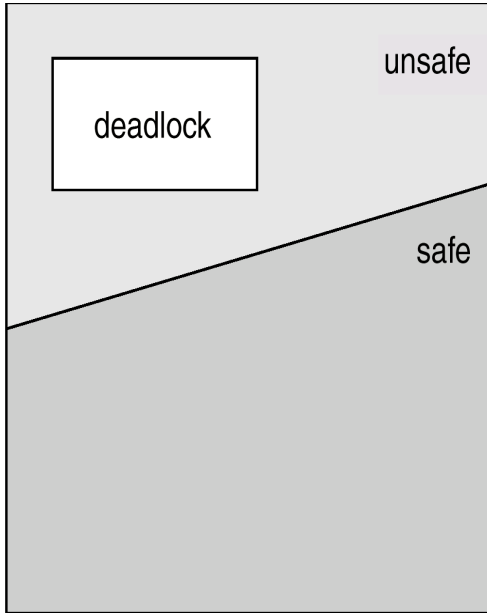
- e.g., CPU, memory space, I/O devices, files
- each resource type R_i has W_i instances.

Resource-Allocation Bipartite Graph $G(V,E)$

- nodes:
 - $P = \{P_1, P_2, \dots, P_n\}$ the set of **processes**
 - $R = \{R_1, R_2, \dots, R_m\}$ the set of **resources types**
- edges:
 - **request edge**: $P_i \rightarrow R_j$
 - If dotted: potential request
 - **assignment edge**: $R_j \rightarrow P_i$



Deadlock avoidance using the OS as arbitrator

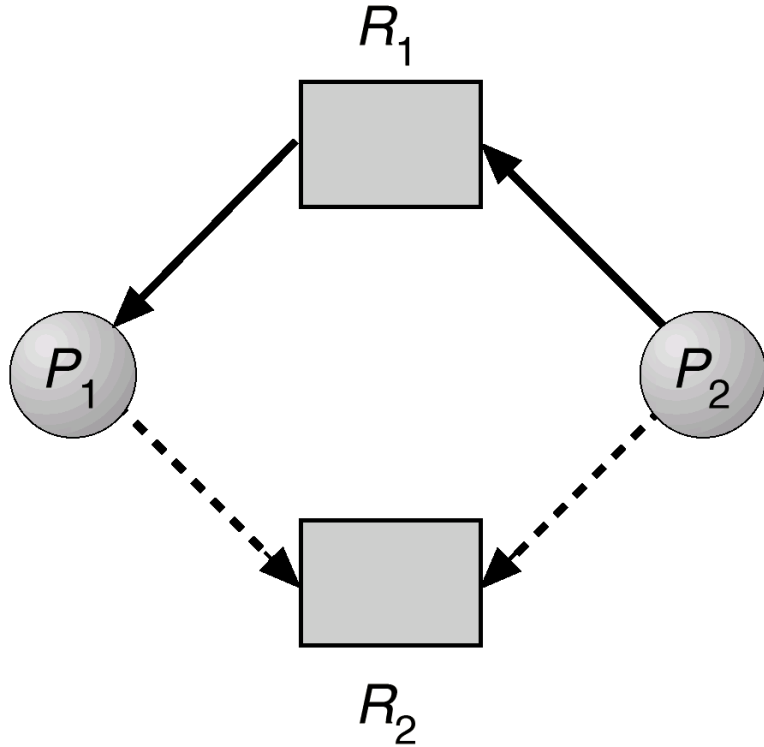


Deadlock-avoidance algo, run by OS:

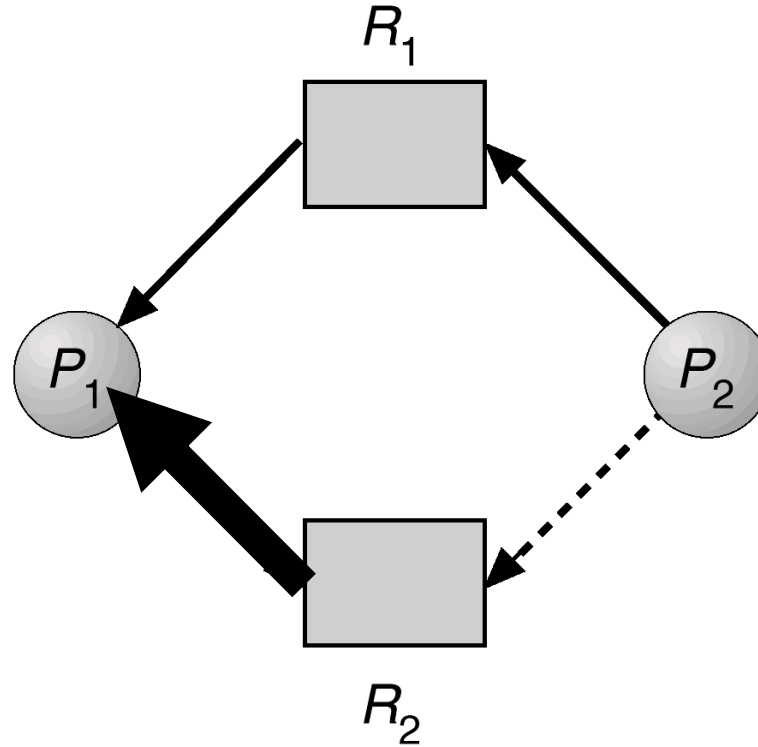
- examines the **resource-allocation state...**
 - *Available, allocated resources*
 - *maximum possible demands* of the processes.
- ...to **ensure there is no potential deadlock:**
 - **unsafe** state $\Rightarrow$ **deadlock might** occur (i.e., later, if all procs request their maximum and no-one can be granted)
- **Avoidance** = ensure that system will not enter an *unsafe* state, by **suspending processes with risky requests**, until enough resources are freed.



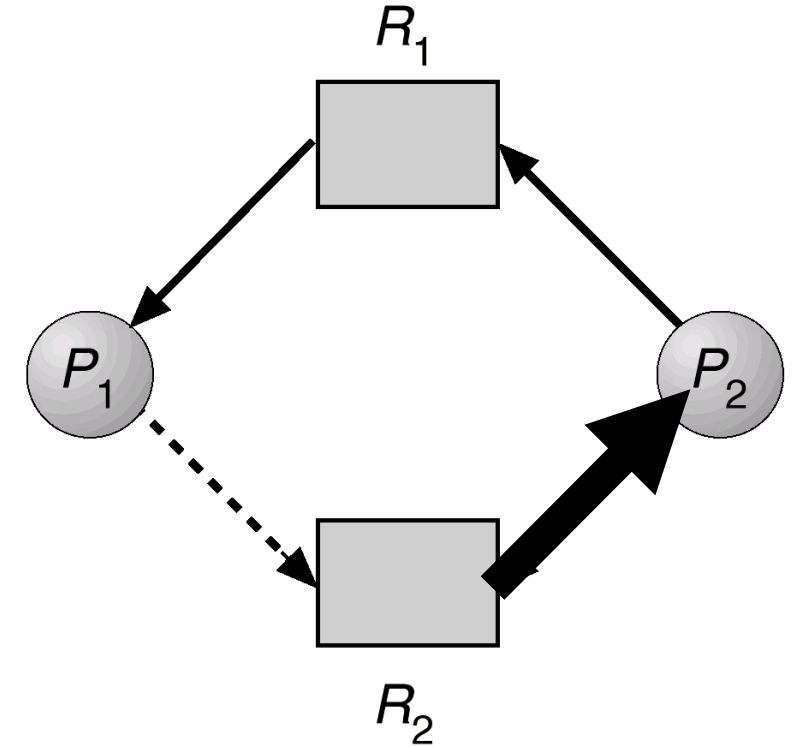
Deadlock avoidance by the OS: example



Example Safe State

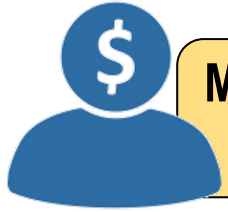


Safe State



Example **Unsafe State**:
i.e., granting the P_2 - R_2 request should not be made until P_1 finishes

Deadlock avoidance by the OS: banker's algorithm



Max [i,j] = k:

P_i may request max k instances of R_j .

Allocation[i,j] = k:

P_i holds k instances of R_j

Available [j] = k:

k instances of R_j are still available.

Avoidance = ensure that system will not enter an unsafe state.

Idea:

If *potentially satisfying a request can* result in an *unsafe state* // i.e., bank will have not enough to let its customers finish and return their loans in case someone requests its max needs

then the *requesting process is suspended*

until enough resources are free-ed // by processes that will terminate in the meanwhile

How to do the safety check efficiently?

Banker's algo gives criterion that can be checked in linear time using the Max, Allocation, Available matrices

Deadlock detection and recovery

- If OS grants requests without checking safety upon every request
- It can allow a deadlock state and when detected (e.g., through detection of cyclical waits), *recover*

(1) Process Termination: Abort all or some deadlocked processes until deadlock is eliminated.

(2) Resource Preemption: Select victim and rollback – return to some safe state, restart process from that state

Must decide on selection criteria (cost, starvation risks, ...)

Recovery is pretty expensive as a method